

Final Project Guidelines

Stock Gap Prediction

Fintech Workshop | Reichman University

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Project Overview

In this project, all groups work on the same question using the same methodology. The differentiator is the quality of your analysis, the external data you find, and your evaluation.

The Question

Will the stock rise significantly tomorrow?

We define "significantly" as a 1-day return greater than 1%. As explained in class, defining the target variable, what counts as a significant move is one of the most consequential decisions you'll make.

Why not just use $\text{return} > 0\%$? A stock closing at $+0.1\%$ would count as positive, even though that move is indistinguishable from rounding error or random fluctuation. The model would treat it as a signal and try to learn from it, which adds noise to the training data and dilutes the patterns we actually care about. Raising the threshold to $+1\%$ ensures every positive observation reflects a genuinely meaningful move.

That said, you are free to adjust the threshold for your sector. Different sectors have different baseline volatility: a 1% daily move in Energy might be significant, while in Technology it could be routine. If you can justify a different threshold based on your sector's behavior, use it.

The Setup

- Each group is assigned one sector from the S&P 500.
- All groups use the same base dataset: 4 years of daily OHLCV data from Polygon.io, plus VIX and S&P 500 returns from yfinance.
- Engineer features from the base data. OHLCV, VIX and S&P return is your raw data, you can derive moving averages, rolling volatility, RSI, returns over different windows, volume ratios, and more.
- Find external data that is specific to your sector. Energy → oil prices. Real Estate → interest rates. What matters for your industry?

Team Size

4 students per group.

Sectors

- Communication Services
- Consumer Discretionary
- Consumer Staples
- Energy
- Financials
- Health Care
- Industrials
- Information_Technology
- Materials
- Real_Estate
- Utilities

Deadline

Each stage is submitted separately throughout the semester. By the final workshop, your project will be essentially complete with only minor refinements needed.

Working with AI

You are expected to use AI tools (Claude, ChatGPT, or others) as your coding partner throughout the project. The AI handles the code. You handle the thinking.

- **Describe.** Tell the AI what you want to understand, not which function to call.
- **Question everything.** The AI will produce plots and numbers. Your job is to ask: does this make sense? What's missing?
- **Catch what AI misses.** The AI might not notice class imbalance, suspicious correlations, or data leakage. You must.
- **Verify independently.** Check ticker counts, date ranges, and target distributions yourself. Don't trust the AI's assumptions.

Deliverables

1. **Jupyter Notebook (.ipynb).** The complete data science workflow in python. Include all stages: data collection, EDA, preprocessing, feature engineering, modeling, and evaluation.
2. **Final Working Paper (PDF).** A compiled document covering all phases. You will build this incrementally, each workshop produces a written section covering that stage. By the end of the course, you compile them into one paper.
3. **Presentation (PowerPoint).** Summarize your project, the problem, analysis, modeling, and key results. Present during the final class session. Focus on the story: what you found, what surprised you, what the model learned (and didn't learn).

Project Workflow

The project follows the five-stage data science cycle. Each workshop focuses on one stage.

Stage 1: Business Problem & Data Collection

- Define the target variable
- Describe your dataset, the base features, the computed features and the external data you add.
- Verify the data is correct, (e.g. ticker count, complete date ranges)
- **Submit:** Code + written section explaining your target variable and data decisions.

Stage 2 + 3:

Stages 2 and 3 are iterative. Preprocessing often reveals new questions that send you back to EDA, and new features open up new patterns to analyze.

Exploratory Data Analysis

- Examine shape, types, and basic statistics.
- Analyze target distribution.
- Univariate distributions, correlation heatmap, feature vs target analysis.
- Identify suspicious features, missing values, outliers.
- **Submit:** Code with EDA visualizations + written section summarizing findings.

Data Preprocessing & Feature Engineering

- Engineer features: e.g. rolling volatility (5/10/20 day), RSI etc
- Find and integrate sector-specific external data.
- Clean the data for modeling
- Handle missing values and outliers
- **Submit:** notebook with feature engineering code + written section explaining feature choices and external data.

Stage 4: Modeling + Evaluation

- Train two models: LightGBM (primary) and Logistic Regression (baseline comparison).
- Use walk-forward evaluation.
- Report average evaluation metrics across all walk-forward steps.
- Choose evaluation metrics that fit your problem and explain why.
- Identify top features. Do they make sense for your sector?
- Explain the model performances, were you able to beat the market?
- **Submit:** notebook with modeling and evaluation + written section justifying model choice, metric selection, and business insights.

Key Rules

Data Splitting

Do NOT use random train_test_split. Your data is time series. Random split leaks dates across train and test. Use walk-forward evaluation: train on expanding windows, test on the next unseen period.

Look-Ahead Bias

Every feature must use only data available at the moment of prediction. If a feature uses future data, it's leaking and your results will be artificially good.

Expected Results

Don't be alarmed if your results look modest. You're trying to beat the market and the market is efficient. Public data is already priced in..